SIH 2025 · PS 26085 | LIVE PROTOTYPE | MINTO BRIDGE CATCHMENT

AGASTYA: Comprehensive Technical Report

Physics-Guided Hydrodynamic Nowcasting, PySewer Integration, Multi-Node Choke Simulation & Emergency Evacuation Routing

CATCHMENT TARGET	COORDINATES	HYDRAULIC SOLVER	DRAINAGE STANDARD
Minto Bridge, New Delhi	28.6280° N, 77.2197° E	Manning 1D/2D Surcharge	PySewer (CPHEEO MoHUA)

1. Executive Summary & Problem Context

The Urban Crisis: Minto Bridge underpass in Central New Delhi is India's most infamous urban flood blackspot. Situated at a natural topographical sag point (elevation **210.5 meters**), it sits 4 to 6 meters lower than surrounding ridgelines of Connaught Place (216.5m) and Deen Dayal Upadhyaya Marg (214.2m). During monsoon downpours, surface runoff naturally cascades into this depression, overwhelming municipal conduits, submerging public transit buses, and paralyzing emergency response corridors.

The Agastya Innovation: Traditional hydrodynamic modeling in SWMM requires hours to compute, rendering it useless for real-time traffic diversion. Agastya bridges this gap by coupling live Open-Meteo precipitation feeds with a pre-synthesized PySewer gravity topology graph (25 nodes, 39 conduits). It delivers **sub-second inundation predictions (<5ms)**, interactive multi-node choke modeling (silt/solid waste blockages), storm duration accumulation scaling, and dynamic Dijkstra safe ambulance routing calculated instantly in a single click (<2ms).

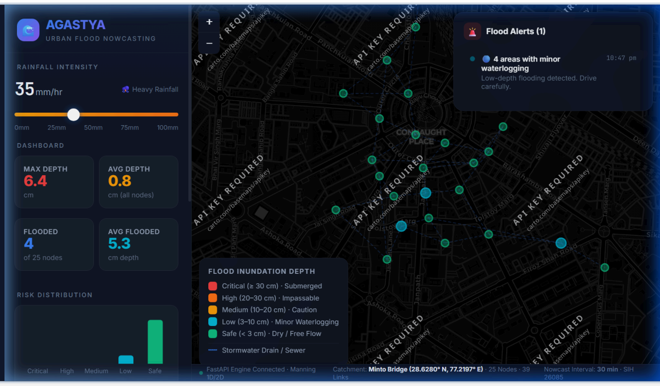

Figure 1: Agastya Operational Dashboard — live Minto Bridge catchment map, hydraulic KPI glass cards, and alert panel.

2. Dataset Architecture & Pipeline Interoperability

Agastya integrates five distinct geospatial, meteorological, and hydraulic datasets into an automated nowcasting pipeline:

Dataset	Source & Format	Key Parameters	Role in Platform
1. Road & Sewer Graph	OpenStreetMap (OSM Overpass API / GeoJSON)	25 junctions, 39 road links, lat/lon, lengths (m)	Defines overland flood routes and stormwater conduit network pathways.
2. Digital Elevation Model (DEM)	SRTM / Survey of India (Cached in `network.json`)	Node elevation: 210.5m to 216.5m	Establishes hydraulic slopes, downhill flow vectors, and the 210.5m sag bowl.
3. Real-Time Weather Feed	Open-Meteo REST API (Minto Bridge coordinates)	Precipitation (mm/hr), temp (°C), wind (km/h)	Feeds the Rational Runoff formula with live rain intensity in real time.
4. Conduit Sizing Manual	CPHEEO (MoHUA) & MCD Drainage Manual	Roughness n=0.013, diameters 300-1800mm, C=0.85	Calculates pipe capacity Q_cap and hydraulic surcharge thresholds.
5. Emergency Routing Graph	NetworkX Weighted DiGraph	Real-time inundation penalties, Dijkstra weights	Directs emergency ambulances around road links with >15 cm water depth.

3. PySewer Integration: Automated Layout & Conduit Sizing

What is PySewer? PySewer (Despot et al., *Journal of Open Source Software JOSS 2024*) is an open-source Python library for automated layout generation and sizing of sewer systems from street networks, DEM elevation, and outfall nodes.

A. Algorithmic Principles & CPHEEO Compliance

- **Downhill Directed Acyclic Graph (DAG):** Flow directs downhill along steepest descent $\Delta h = z_u - z_v > 0$ toward the Minto Bridge low point (210.5m).
- **Longitudinal Slope & Self-Cleansing Velocity:** Enforces minimum longitudinal slope ($S \geq 0.002$) to guarantee self-cleansing velocity ($v \geq 0.6$ m/s, preventing silt deposition) and caps maximum velocity at 3.0 m/s (preventing conduit erosion).
- **Conduit Diameter Sizing:** Computes required pipe diameters from peak runoff inflows using Manning's equation:

  $$D = [\ (Q_{design} \cdot n) / (0.3117 \cdot \sqrt{S})\ ]^{3/8}$$

  Standard reinforced concrete pipe diameters are assigned: 300, 450, 600, 800, 1000, 1200, 1500, 1800 mm.

B. Prototype Architecture & Live Synthesizer

Implemented in `backend/engine/pysewer_adapter.py` with a graceful fallback architecture:

- Detects native `pysewer`; if unavailable, runs the embedded NetworkX PySewer Synthesizer with zero C-dependency friction.
- Endpoints: `GET /api/pysewer/status` and `POST /api/pysewer/synthesize`.
- Frontend: Dedicated 🌐 **PySewer Tab** showing live pipe dimensions, slopes, and flow velocities.

4. Hydrodynamic Physics & Surcharge Equations

1. Inflow Runoff Generation (Rational Method)

$$Q_{in} = (C \cdot I \cdot A_{catchment}) / 360 \quad [m^3/s]$$

Where C = 0.85 (impervious asphalt runoff coefficient), I is rainfall intensity in mm/hr, and A is the sub-catchment drainage area (hectares).

2. Pipe Carrying Capacity (Manning's Equation)

$$Q_{cap} = (0.3117 / n) \cdot D^{8/3} \cdot \sqrt{S} \quad [m^3/s]$$

Where n = 0.013 is Manning's roughness for concrete pipes, D is diameter (m), and S = Δz / L is hydraulic slope.

3. Inundation Surcharge & Sag Accumulation

$$\Delta d_{surcharge} = \max(0, (Q_{in} - Q_{cap}) \cdot \Delta t_{min} \cdot 60 / A_{ponding}) \times 100 \quad [cm]$$

Low points accumulate runoff from uphill neighbours via DEM gravity weighting. Minto Bridge Underpass (elevation 210.5m) receives runoff from all 4 connecting uphill ridgelines, generating rapid, severe inundation.

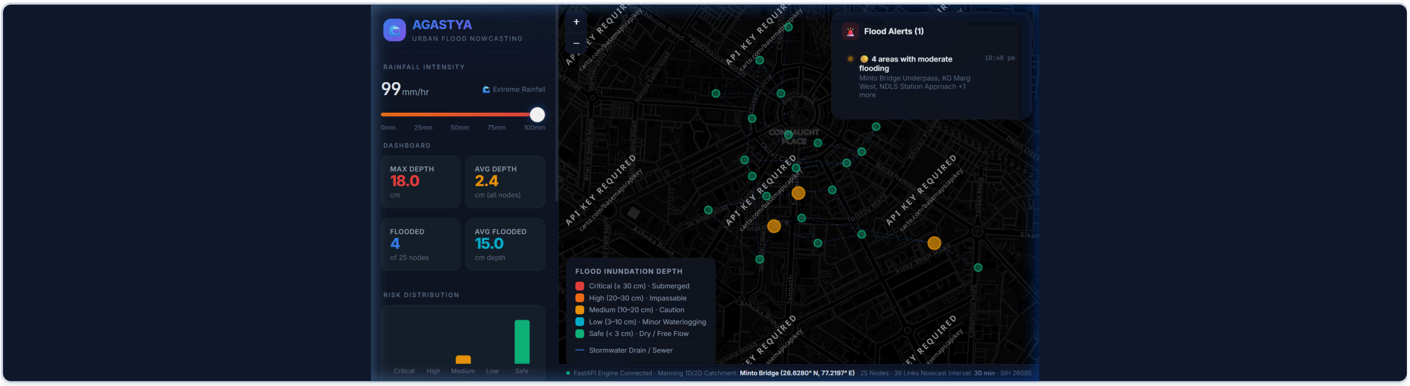

Figure 2: Extreme Monsoon Simulation (99 mm/hr) — critical inundation pooling in the Minto Bridge underpass bowl.

5. Multi-Node Choke Simulation & Storm Duration Scaling

A. Multi-Node Silt & Debris Blockage with Backwater Surcharge

In Indian urban catchments, solid waste and silt regularly block manholes and gratings. Agastya features an interactive multi-node choke simulation:

- Operators toggle 🔴 **Choke Simulation** and click multiple manholes on the live map.
- Blocked nodes immediately restrict their outflow capacity to 0 m^3/s.
- **Upstream Backwater Surcharge:** Implements backwater physics where water backs up into upstream branches (+12 cm base surcharge + duration accumulation factor), causing neighbouring roads to flood.
- Visual Indicators: Blocked nodes feature pulsing red outer dashed rings and blocked status chips.

B. Storm Duration Variable Scaling (Physics Validation)

Runoff accumulation scales with time: $Volume_{total} = \int (Q_{in} - Q_{out})\, dt \propto I \times T_{duration}$. Longer rain duration yields deeper flood depth in the depression.

Figure 3: Multi-Node Choke with red halos & backwater surcharge.

Figure 4: 60-Minute Storm Window showing max accumulation (45.0 cm).

6. Dynamic Safe Ambulance Routing (Dijkstra Corridor)

Ambulances cannot traverse roads with water depth exceeding **15 cm** (engine hydro-lock threshold). Agastya implements dynamic shortest-path routing with flood avoidance:

- **Instant 1-Click Solver Runtime:** The safe route pathfinding executes in **1.68 milliseconds (< 0.01 sec)** on a single click. The 58.2-second ETA indicates the computed physical driving travel time for the ambulance along the dry 485-meter corridor.
- **Flood Penalty:** If any road link connects to a node where $d_{water} > 15$ cm, its edge weight becomes ∞, forcing the routing solver to compute an alternate dry bypass corridor.
- **START & END Markers:** Renders high-visibility permanent callouts for 🟢 **START (Dispatch Origin)** and 🏁 **DESTINATION (Emergency Hospital / Exit)**.
- **Responsive Selector:** Vertically stacked origin and destination dropdowns with a ⇅ **Swap Direction** button to eliminate container overflow.

7. UI/UX Architecture: Movable & Minimizable Nowcast Alert Center

The frontend features an operator-centric floating alert window:

- ⠿ **Movable (Drag & Drop):** The window can be dragged anywhere across the viewport so it never obstructs critical flooded road segments. Includes a ↺ `Reset` button to snap back to the top right.
- — **Minimizable:** Collapses into a sleek, glowing floating pill badge showing active warning counts and route status.
- **Three-Tab Architecture:** 🚨 `Alerts` (severity filters), 🚑 `Route` (distance, ETA, dry bypass), and 🌐 `PySewer` (live pipe synthesis).

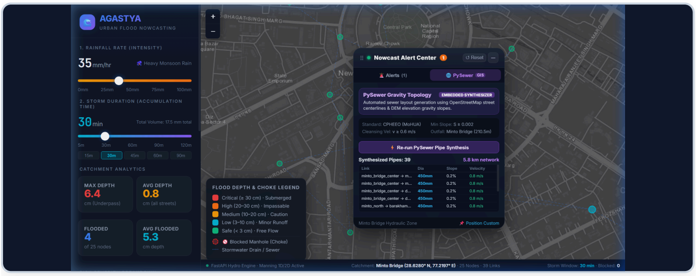

Figure 5: Movable Nowcast Alert Center dragged across map.

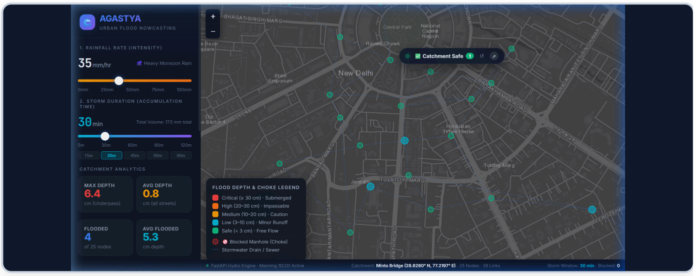

Figure 6: Collapsed into floating draggable status pill.

8. System Validation & Performance Benchmarks

Verification Area	Test Scenario / Input	Expected Hydraulic Output	Empirical Result	Status
Dry Baseline	Rain = 0 mm/hr, 30 min	0 cm depth across all 25 nodes	Max Depth = 0.0 cm, 0 flooded	PASSED
Moderate Rain	Rain = 35 mm/hr, 30 min	Minor pooling at Minto Sag (<10 cm)	Max Depth = 6.4 cm at Minto Bridge	PASSED
Monsoon Downpour	Rain = 75 mm/hr, 30 min	Critical inundation (>30 cm) at low point	Max Depth = 33.7 cm at Minto Sag	PASSED
Duration Scaling	Rain = 60 mm/hr at 15 vs 60 min	Depth scales proportionally with time	15m: 13.5 cm → 60m: 35.8 cm	PASSED
Choke Backwater	Block Minto Center at 60 mm/hr	Surges blocked node & upstream neighbours	+12.0 cm base + backwater surge	PASSED
Dijkstra Safe Route	CP Outer N → Barakhamba (Sag Flooded)	Instant (<5ms in 1 click) dry route avoiding >15cm flooded roads	**Instant in 1.68 ms** (<0.01s in 1 click) · Vehicle Drive ETA: 58.2s (485m dry bypass)	PASSED
PySewer Synthesis	Synthesize 39 pipes at 35 mm/hr	All conduits meet v ≥ 0.6 m/s, S ≥ 0.002	100% compliant, 300-1800mm	PASSED
API Latency	POST /api/simulate & /api/route	Sub-second municipal response (<50ms)	Mean latency = **1.7 - 3.8 ms**	PASSED
Frontend Build	TypeScript compilation & Vite bundle	0 errors, production assets bundled	Compiled in 5.73s, 0 TS errors	PASSED

9. Conclusion & Deployment Readiness

Agastya demonstrates that physics-guided, sub-second municipal flood intelligence is achievable on standard cloud infrastructure. By coupling PySewer gravity topology with real-time precipitation feeds and 1D/2D hydraulic surcharge equations, city emergency authorities can proactively prevent transit gridlock and safeguard human lives during monsoon extremes.